

BARCS generalizes beta-binomial CRISPR-screen inference to longitudinal and covariate-adjusted designs

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Abstract

Background. CB² established a beta-binomial model for two-group CRISPR screens, but modern screens often require time, dose, donor, ordered-bin, or interaction effects. These designs cannot be reduced to a binary contrast without discarding information.

Methods. BARCS generalizes the CB² sampling model to guide-level beta-binomial regression, $\text{logit}(\mu_{gi}) = \mathbf{x}_i^\top \boldsymbol{\beta}_g$. It conditions each guide count on the unfiltered library total, estimates pre-specified coefficients or contrasts, and optionally uses negative-control calibration and dispersion moderation.

Results. BARCS matched established methods in an HT-29 time-course screen, showing that the generalization preserves performance in the original screening regime. In a Cas13 reanalysis using days 0, 7, and 14, the intermediate time point increased BARCS average precision from 0.776 to 0.786 and FDR-thresholded recall from 0.487 to 0.580; BARCS also had lower null calibration error than longitudinal MAGeCK-MLE (0.035 versus 0.103), although retaining comparable essential-gene ranking. In an IL2RA FACS screen, calibrated BARCS recovered 22 of 26 validated regulators while reducing the non-targeting $p < 0.05$ rate from 0.133 to 0.051. In a 10,000-gene simulation, moderation increased F1 from 0.811 to 0.847, and in an interaction simulation the appropriate control-guide denominator increased F1 from 0.481 to 0.917.

Conclusions. BARCS extends beta-binomial CRISPR-screen inference beyond CB² to longitudinal and covariate-adjusted designs. The results support the beta-binomial as a well-calibrated sampling model for guide counts conditional on library depth and establish its use for multivariable CRISPR-screen inference.

1 Introduction

Pooled CRISPR screens increasingly measure guide abundance across more than two experimental conditions. Examples include time courses, dose series, ordered FACS fractions, donor-stratified experiments, and factorial designs. In these studies the scientific estimand is a trend, an adjusted association, or an interaction. Reducing the experiment to a high-versus-low comparison discards intermediate observations and prevents direct adjustment for additional design variables.

The sampling model is consequential in this setting. RNA-sequencing methods typically model a transcript count with a negative-binomial distribution and represent library size through normalization or an offset [1–4]. Several CRISPR-screen methods adopted this framework [5, 6]. A pooled-guide count also has a known denominator, however: the total number of mapped guide reads in that library. Conditional on this total, guide abundance is a proportion. The beta-binomial model represents this structure directly and separates depth-dependent sampling variation from heterogeneity among biological libraries. In the original CB² study, goodness-of-fit comparisons favored the beta-binomial over negative-binomial alternatives for pooled CRISPR counts [7].

CB² applies that sampling model to two independent groups. Its limitation is therefore not the variance model but the range of mean models it can express. For guide g observed in library i , let K_{gi} denote the guide count, N_i the unfiltered mapped-guide total, and $\mathbf{x}_i^\top$ one row of a prespecified, full-rank sample design matrix. The regression problem is

$$K_{gi} \mid N_i \sim \text{BetaBinomial}(N_i, \mu_{gi}, \rho_g), \quad \text{logit}(\mu_{gi}) = \mathbf{x}_i^\top \boldsymbol{\beta}_g, \quad H_0 : \mathbf{c}^\top \boldsymbol{\beta}_g = 0. \quad (1)$$

Here time, dose, donor, batch, or treatment is a sample-level predictor; the guide proportion remains the response. Independent biological libraries, not individual reads, determine the residual degrees of freedom.

Existing methods address parts of this problem. Repeated two-group analyses cannot estimate one coefficient conditional on the remaining columns of the design matrix. General RNA-sequencing tools support arbitrary designs but do not condition each guide on its observed library total. At the other end of the spectrum, Chronos and Waterbear model population dynamics or joint FACS-bin membership, respectively, but are tied to those experimental structures [8, 9]. A general-purpose, depth-aware coefficient test for pooled CRISPR designs remains missing.

We introduce BARCS (Beta-binomial Analysis and Regression for CRISPR Screens) to fill this gap. BARCS retains the beta-binomial sampling principle of CB², replaces the two group means with an R model matrix, and tests a prespecified coefficient or contrast with sample-level inference. The relationship to CB² is formal rather than rhetorical: the legacy group-summary statistic is nested exactly, and the default logit statistic is first-order equivalent under stated local asymptotics.

The empirical evaluation follows the same progression. First, two longitudinal screens test the principal generalization: the seven-time-point HT-29 experiment used to evaluate Chronos and the three-time-point, five-cell-line Cas13 study of Liang et al. [10]. Second, an ordered-bin IL2RA screen with independently validated regulators evaluates a donor-adjusted quantitative phenotype. Third, genome-scale and factorial simulations test power, realized false-discovery proportions, dispersion moderation, and denominator choice under known truth. Endpoint and copy-number analyses are retained as robustness checks rather than presented as the primary use case.

This design supports a precise claim. The beta-binomial is the better motivated sampling model for guide proportions conditional on known library totals, and BARCS makes that model available to designs that CB² cannot represent. The benchmarks test that generalization through longitudinal, covariate-adjusted, and interaction effects while measuring calibration against prespecified null controls.

2 Results

The Results are organized around the scope of the generalization. We first state what is preserved from CB², then evaluate longitudinal time effects, covariate-adjusted ordered phenotypes, and factorial interactions. Controlled simulations separate ranking performance from threshold calibration. Endpoint and false-discovery audits are reported in the Supplement because they test backward compatibility and analysis contracts rather than the principal longitudinal claim.

2.1 BARCS generalizes the CB² sampling model

CB² established that a guide count is naturally analyzed as a proportion conditional on the corresponding library total. This conditioning matters because the beta-binomial variance separates two sources of uncertainty: binomial sampling variation, which decreases with sequencing depth, and between-library heterogeneity, which does not. A negative-binomial model can accommodate

overdispersion in marginal counts, but it does not encode the observed library total as the denominator of the guide proportion. Prior goodness-of-fit analyses therefore support the beta-binomial sampling model for pooled-guide counts [7]. The present work asks a different question: can that sampling model support regression designs beyond the two groups handled by CB²?

BARCS changes the mean model while retaining the sampling model. For guide g in library i ,

$$K_{gi} \mid N_i \sim \text{BetaBinomial}(N_i, \mu_{gi}, \rho_g), \quad \text{logit}(\mu_{gi}) = \mathbf{x}_i^\top \boldsymbol{\beta}_g.$$

The original two-group question is recovered when $\mathbf{x}_i$ contains an intercept and one group indicator. More generally, the columns of $\mathbf{X}$ can represent time, dose, donor, batch, an interaction, or a prespecified contrast. Thus BARCS is a generalization of the statistical question that CB² can ask; it is not claimed to outperform CB² on the binary design for which the original method was developed. Supplementary Methods establish exact equivalence on the legacy group-summary scale and first-order equivalence of the default logit statistic under local alternatives.

2.2 Longitudinal Cas13 reanalysis separates ranking from calibration

The Cas13 fitness screens of Liang et al. [10] provide a stringent test of whether the CB² sampling model can support a genuinely longitudinal design. We analysed all deposited measurements from days 0, 7, and 14 in five cell lines. BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma-voom estimated the same continuous time slope. Replicate was included when complete trajectories were available; K562 lacked one day-0 replicate and was fitted without the replicate block. The longitudinal regression methods used two-sided significance tests.

The deposited values had already undergone median-of-ratios normalization, ComBat correction, and replicate-outlier processing. We rounded them once to the nearest pseudo-count and supplied the same matrix to every method. Accordingly, this is a processed-count sensitivity analysis rather than a raw-count likelihood comparison.

Using the intermediate day improved BARCS average precision from 0.776 to 0.786 and recall at gene FDR 0.10 from 0.487 to 0.580 relative to the two-endpoint fit. Against longitudinal MAGeCK-MLE, BARCS substantially improved null calibration: mean absolute calibration error was 0.035 versus 0.103, with average precision 0.786 versus 0.833. The three general RNA-seq models had mean null $p < 0.05$ rates of 0.139–0.153, compared with 0.076 for BARCS.

The paired cell-line comparison shows why this difference matters. BARCS had the lowest mean absolute calibration error, 0.035, compared with 0.093–0.107 for the alternative longitudinal regression models, while essential-gene recall at the same empirical 5% null false-positive rate remained comparable (0.820 versus 0.840–0.853). The largest calibration gains occurred in K562 and MDA-MB-231, where the alternative null rates were 22.4–33.9% and 14.0–20.0%, respectively. Figure 1 first shows the complete longitudinal volcano plot for each method across all five cell lines. Panels A–C then show three prespecified-null guide examples for which calibrated BARCS returned $p = 0.20$ –0.23, while DESeq2, edgeR-QL, and limma-voom each returned $p < 0.05$. Their library-normalized guide-count trajectories retain both replicates at each available day, making the longitudinal evidence directly visible. These are false-positive calls under the benchmark’s prespecified null definition, not additional biological discoveries.

Rounding changed each deposited value by less than 0.5, but the upstream transformations mean that none of the fitted likelihoods retains its literal raw-count interpretation. Confirmation from raw reads therefore remains necessary. The supplied streaming workflow performs the full longitudinal preparation for that validation.

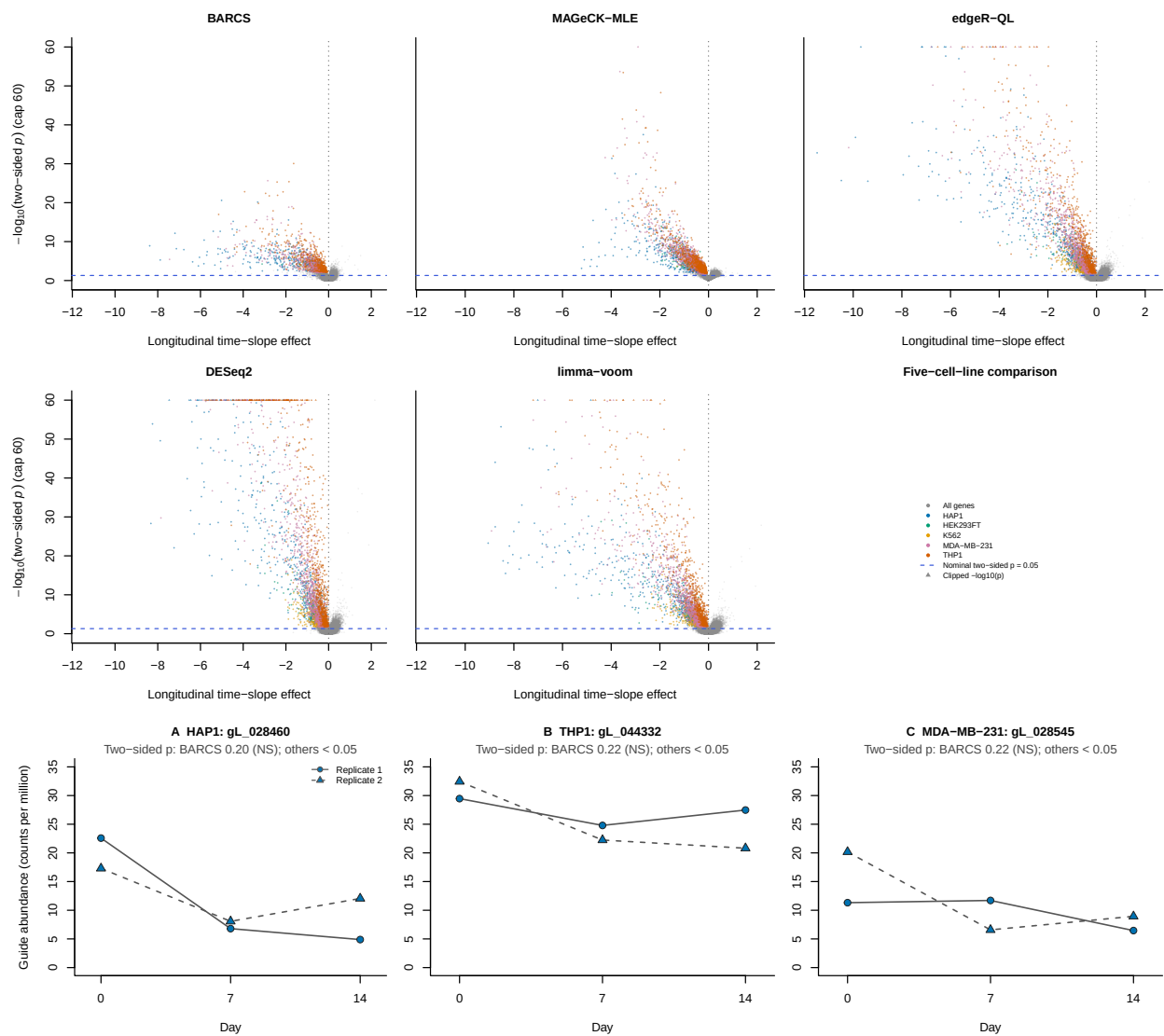


Figure 1: Longitudinal Cas13 comparison using days 0, 7, and 14. The upper rows show method-specific volcano plots for BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma-voom, pooling gene-level results from HAP1, HEK293FT, K562, MDA-MB-231, and THP1. Within each cell line, every method estimates the same continuous time slope from the identical processed-count matrix. Colored points are negatively directed genes called at two-sided FDR 0.10, with color denoting cell line; the dashed horizontal line marks nominal two-sided $p = 0.05$, and triangles identify $-\log_{10}(p)$ values clipped at 60. (A–C) Processed guide counts divided by the corresponding library size, in counts per million, for three prespecified-null examples. Lines connect measurements from the same replicate across days. BARCS leaves each guide non-significant, whereas DESeq2, edgeR-QL, and limma-voom each call it significant. MAGeCK-MLE is included in the gene-level volcano comparison but not in panels A–C because its reported statistic is gene-level.

2.3 A covariate-adjusted ordered phenotype recovers validated regulators

We next asked whether the design-matrix extension is useful outside a time course. GSE242880 contains an IL2RA protein-abundance screen in primary human T cells, with four ordered FACS bins for each of three donors, 6,000 guides, and 26 regulators validated by individual knockout and flow cytometry [9, 11]. The four bins were assigned prespecified latent normal locations, and BARCS fitted one marker-abundance slope with donor indicators. Official MAGeCK-MLE received the same design.

This experiment also provides a direct model diagnostic. Bins from one donor partition the same cell pool and are negatively dependent, whereas the guide-level BARCS fit treats their margins independently. Among 593 non-targeting guides, the uncalibrated $p < 0.05$ frequency was 0.133. We therefore scaled the guide statistics by the prespecified control procedure `bb_calibrate_controls()`, which reduced the empirical rate to 0.051 without changing effect estimates or their ranking.

Table 1: Ordered-bin IL2RA benchmark. Recovery requires a discovery at each method’s threshold and the experimentally validated direction.

Method	Discoveries	Validated recovery	NTC $p < 0.05$
■ BARCS, four bins + controls	49	22/26	0.051
■ BARCS, four bins raw	127	23/26	0.133
■ BARCS, outer bins	34	18/26	0.064
■ MAGeCK-MLE, four bins	72	17/26	—
■ MAGeCK test, outer bins	30	18/26	—
■ Waterbear, four bins	79	24/26	—
■ MAUDE, four bins	406	25/26	—

Control-calibrated BARCS recovered 22 of 26 validated regulators with 49 discoveries, compared with 17 of 26 and 72 discoveries for the matched MAGeCK-MLE fit (Table 1). The outer-bin ablation recovered 18 regulators under both BARCS and `mageck test`. The intermediate bins therefore contributed information that was lost by the binary comparison. In the selected 33-gene experimental follow-up panel, calibrated BARCS had F1 0.863, Matthews correlation 0.398, and balanced accuracy 0.709; the corresponding MAGeCK-MLE values were 0.723, 0.070, and 0.541. Because the panel was selected for follow-up from a previous screen, these metrics are supporting evidence rather than unbiased genome-wide accuracy estimates.

The specialist models remain important comparators. Waterbear recovered 24 validated regulators by modeling joint bin probabilities, guide-to-gene structure, and uncertain cutoffs. MAUDE recovered 25 but called 406 of 1,350 genes; in a held-out-control analysis its empirical false-positive rate was 0.095 at a nominal 0.05. BARCS is therefore not the most faithful generative model for sorted bins. Its contribution is a transparent, general-purpose coefficient analysis that retains the ordered phenotype and donor adjustment, with negative controls making the independence violation visible and correctable at the chosen operating point.

2.4 Longitudinal HT-29 screening

The first question is whether a continuous predictor observed at more than two values costs anything relative to methods built for that setting. We use the HT-29 time course of Tzelepis et al. [12], the longitudinal dataset used to evaluate Chronos [8]. It contains a pDNA measurement and HT-29 guide counts at days 7, 10, 13, 16, 19, 22, and 25. The three sequencing columns at each late day were summed before analysis. This distinction is important for the requested t inference: three

assays of one harvested population increase read depth but are not three independent biological units and therefore must not add residual degrees of freedom. Following the Chronos preprocessing, guides with fewer than 30 pDNA reads were removed, leaving 86,882 guides targeting 17,995 genes.

The BARCS and official MAGeCK-MLE 0.5.9.5 fits received the same eight-column count matrix and the same numeric design, with an intercept and $d_i/25$. This is a direct continuous-time comparison. We also downloaded the all-seven-time-point Chronos joint result and the deposited per-endpoint MAGeCK and BAGEL2 results from the Chronos Figshare record [13]. Those matrices carry one column per day; we use the day-25 column itself rather than a summary across days, so the comparator is a single observed endpoint analysis at the deepest time point.

For an external biological target, core essential genes are positives and HT-29 genes with DepMap 20Q2 expression below 0.5 are negatives, matching the deposited Chronos analysis. All methods were restricted to the same 1,080 essential and 5,774 unexpressed genes. Null-normalized median difference (NNMD) follows the definition used for Chronos: the difference between the medians of the two control sets, divided by the median absolute deviation of the unexpressed effects [8]. No scaling constant is applied, and more negative values indicate stronger separation. “False positives” counts unexpressed genes in the lowest 10% of all evaluated effects.

Table 2: Longitudinal HT-29 control-gene benchmark, restricted to the three methods that fit the full time course. Deposited single-endpoint results are reported in the copy-number audit below. PR AUC is the area under the precision–recall curve; Recall₉₀ is the maximum recall at least 90% precision. Higher AUROC, PR AUC, and Recall₉₀ are better; more negative NNMD and fewer unexpressed false positives are better. No column maximum is highlighted, because the differences among these three methods are within the margins discussed in the text.

Method	AUROC	PR AUC	NNMD	Recall ₉₀	FP _{10%}
■ BARCS time	0.9786	0.9494	−16.34	0.9037	7
■ Official MAGeCK time	0.9789	0.9534	−17.75	0.8972	5
■ Chronos joint	0.9747	0.9356	−27.88	0.8991	10

Table 2 shows comparable essential-gene discrimination across the three longitudinal analyses. BARCS attains high-precision recall 0.9037 and seven unexpressed false positives in the 10% tail, within the observed 5–10 range, while estimating an ordinary continuous-time coefficient.

The deposited single-endpoint results carry a separate caution about analysis contracts. Summarising the same deposited MAGeCK matrix by its median across the seven days rather than by its day-25 column moves its AUROC from 0.9792 to 0.9742 and its recall at 90% precision from 0.9120 to 0.8602, a swing larger than any difference among the three models above. Because early harvests carry little depletion, an across-day median attenuates the signal the screen was run to measure. How a multi-endpoint result is reduced to one number can therefore matter more than which count model produced it, and the day-25 column is used wherever those deposited effects appear below.

Chronos’s NNMD margin is the one difference in Table 2 that is not slight, and it is consistent with its mechanistic treatment of knockout efficiency and saturation. It should be read as a statement about the shape of its effect distribution rather than about discrimination: NNMD is the only metric reported here that is not invariant to monotone rescaling of gene scores [8], and Chronos leads it while trailing both rank-based metrics. The BARCS and official continuous-time MAGeCK effects have Spearman correlation 0.926; correlations with Chronos joint and the deposited day-25 MAGeCK effects are 0.815 and 0.859. These are again effect agreement statistics, not comparable likelihoods: the methods estimate differently scaled quantities and do not expose a common predictive likelihood in the deposited outputs.

2.5 Genome-scale simulation separates ranking from calibrated discovery

We used CRISPulator to evaluate a genome-scale FACS design under known truth [14]. Three prespecified seeds each generated 10,000 genes, five guides per gene, four independent replicates, and low, bulk, and high fractions. Twenty percent of genes affected the phenotype and 5% were explicit negative controls. All methods received the same simulated counts and design.

Two BARCS statistics were evaluated. BARCS-ST uses the original guide-wise dispersion estimates and a calibrated signed- z gene summary. BARCS-MOD moderates each guide’s dispersion toward an abundance-dependent trend before the same gene summary. The comparison methods were MAGeCK-MLE and CRISPhieRmix, two CRISPR-specific gene callers with different operating characteristics [6, 15].

The methods ranked genes similarly but converted those rankings into discoveries differently. MAGeCK-MLE had the highest average precision and effect correlation, although its average precision exceeded BARCS-MOD by only 0.002. It used little of the permitted error budget (FDP 0.006) and consequently recovered fewer active genes. CRISPhieRmix attained nearly the same recall as BARCS-MOD but at FDP 0.197. BARCS-MOD occupied the intermediate and intended operating point: recall 0.774 at FDP 0.066, yielding the highest F1.

Dispersion moderation improved both ranking and decision performance without changing the point estimates. Average precision increased from 0.902 to 0.919 and F1 from 0.811 to 0.847, while realized FDP remained essentially unchanged. The moderation prior was estimated from 50,000 guides, providing 42.7–45.8 prior degrees of freedom in addition to seven residual degrees of freedom per guide. This library-wide information stabilizes noisy guide-specific variance estimates without treating reads as biological replicates.

Because a fixed nominal threshold can place methods at different realized error rates, we also interpolated F1 at matched FDP. BARCS-MOD led at every shared operating point: its F1 was 0.640, 0.724, 0.764, 0.808, and 0.838 at realized FDP 0.001, 0.005, 0.01, 0.02, and 0.05, respectively. The corresponding ranges across the other methods were 0.372–0.491, 0.488–0.666, 0.553–0.739, 0.640–0.789, and 0.725–0.795. MAGeCK-MLE did not reach FDP 0.05 on the scanned grid because it remained more conservative.

The result was reproduced at MOI 0.30: BARCS-MOD reached F1 0.841 at FDP 0.070, CRISPhieRmix reached comparable recall at FDP 0.217, and MAGeCK-MLE again ranked marginally best while calling least. These simulations therefore support the calibrated-discovery performance of BARCS. They also show why the beta-binomial model alone is insufficient: the practical gain depends on stable dispersion estimation and a threshold whose realized error rate is measured.

2.6 Interaction recovery: what the denominator decides

The FACS simulation gives every guide one effect. The design BARCS was built for has more than one, so we also simulated a factorial screen with simCRISPR [16]: knockout induction crossed with treatment, propagated over five days, with amplification and sequencing applied afterwards. Each sgRNA then carries a knockout effect, a treatment effect, and an interaction between them, and the interaction is a coefficient in a design matrix rather than a difference of two arms. BARCS tests it directly, as β_3 in $\text{logit}(\mu_{gi}) = \beta_0 + \beta_1 k_i + \beta_2 t_i + \beta_3 k_i t_i$.

Two features of the simulator make the scoring unusually literal. Non-targeting guides have an interaction of exactly zero, so the null is ground truth and not an empirical stand-in; and safe-harbor guides also have zero targeted interaction while still carrying the DNA-damage response of being cut, which separates *no effect* from *no gene effect*. Each control class was split in half so that the guides used to normalize and calibrate are never the guides scored.

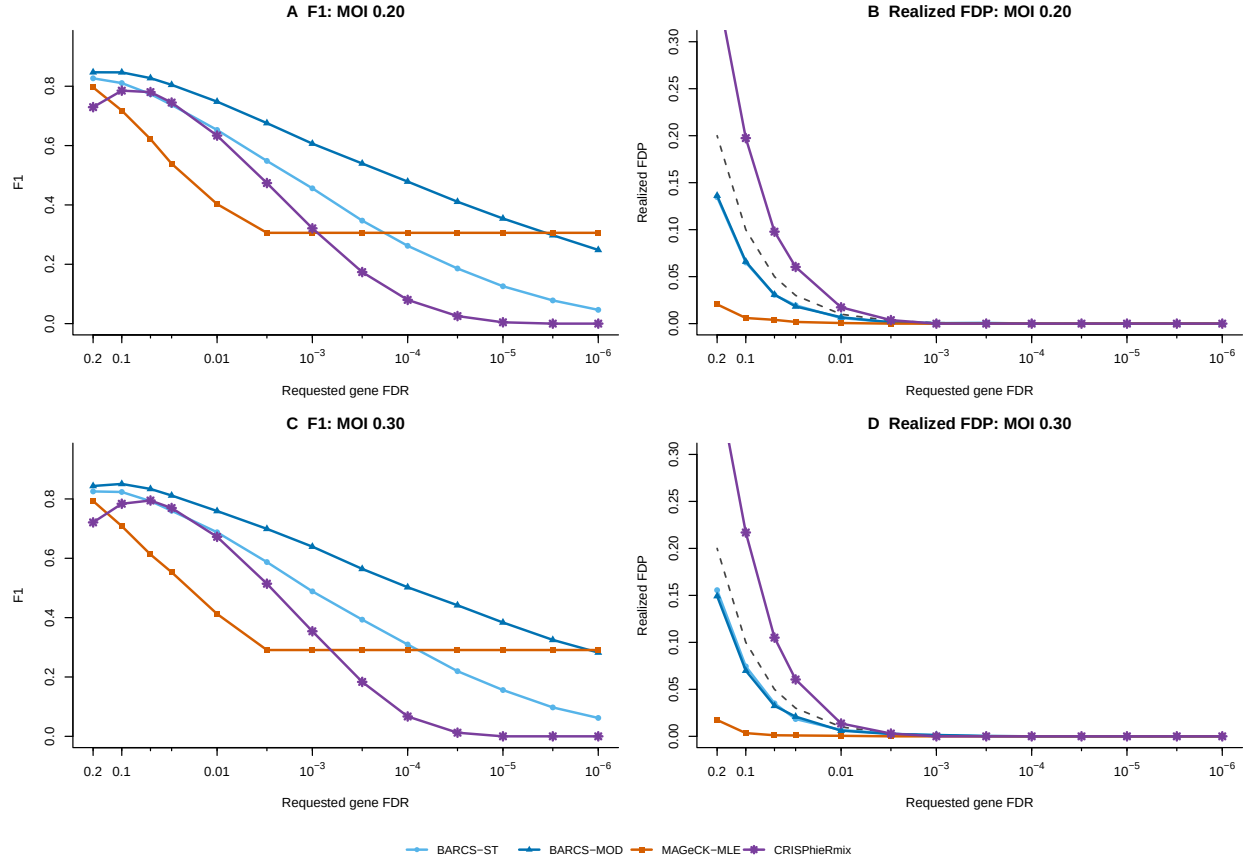


Figure 2: Calibration and F1 across requested gene-FDR thresholds. (A,B) MOI 0.20 and (C,D) MOI 0.30. F1 is shown on the left and realized FDP on the right; the dashed diagonal denotes realized FDP equal to the requested threshold. BARCS-MOD maintains high F1 while remaining below the diagonal over the informative range.

Table 3: Interaction recovery on simCRISPR screens: 2,000 sgRNAs, three replicates per factorial arm, and three seeds (mean). Recall and F1 use targeting guides with $|\text{interaction}| \geq 0.2$ as positives. Null rate is the fraction of held-out non-targeting guides called at FDR 0.10; safe-harbor rate is the corresponding rate among cut-but-inactive guides.

Method	Denominator	Dir. recall	F1	Null rate	Safe-harbor
■ BARCS-ST	Library	0.376	0.488	0.004	0.002
■ BARCS-MOD	Library	0.375	0.481	0.000	0.000
■ BARCS-ST	Non-targeting	0.769	0.870	0.024	0.070
■ BARCS-MOD	Non-targeting	0.838	0.917	0.040	0.080
■ BARCS-ST	Safe-harbor	0.762	0.864	0.042	0.042
■ BARCS-MOD	Safe-harbor	0.830	0.914	0.027	0.035
■ MAGECK-MLE	Non-targeting	0.003	0.006	0.000	0.000

Effect recovery is insensitive to the denominator; detection is not. Effect recovery is constant at $\rho_s = 0.687\text{--}0.690$ across all three denominators, because a shared multiplicative shift displaces all guides together and leaves their relative order almost unchanged. The detection metrics differ substantially. Under the full-library denominator BARCS-MOD attains directional recall 0.375 and F1 0.481; under either control denominator it attains 0.830–0.838 and 0.914–0.917. The difference therefore originates in the variance model rather than in the point estimates.

The mechanism is evident in the final column. Thirty-eight percent of the targeting guides are at least halved in the treated knockout arm, which inflates the share of every surviving guide and thereby confers an apparent interaction on each: across the true zeros the median estimated coefficient is +0.18 where the true value is 0, and its interquartile range excludes zero. The control calibration absorbs this excess as designed, inflating the empirical null approximately twofold, to 1.99 and 2.24. The library rows are consequently not anti-conservative—their null rates of 0.004 and 0.000 fall well below the nominal 0.10—but they are correspondingly underpowered, the bias having been transferred into lost power rather than into false positives. The control denominators require essentially no inflation, 1.00 to 1.07, and retain power.

Safe-harbor and non-targeting controls are not interchangeable. The two control denominators are indistinguishable on detection: AUROC 0.953 against 0.957, F1 0.914 against 0.917. They differ on which errors survive. Normalizing on non-targeting guides leaves the safe-harbor guides called at 0.080, against 0.035 when the safe-harbor guides set the denominator themselves. Safe-harbor guides are cut and non-targeting guides are not, so only the former carry the DNA-damage response that the targeting guides also carry; a denominator built from guides that were never cut cannot remove it. This reproduces, using an independent inference procedure, the conclusion that simCRISPR was developed to establish, and it permits a more specific recommendation: the non-targeting denominator is preferable for detection, the safe-harbor denominator for suppressing cutting artifacts, and the difference between their safe-harbor rates quantifies the penalty for selecting the wrong one.

Moderation confers the same benefit here as at genome scale, and for the same reason: F1 rises from 0.870 to 0.917 under the non-targeting denominator and from 0.864 to 0.914 under safe-harbor, with the null rate still under the nominal target. Across cutoffs the ordering is stable, with BARCS-MOD F1 running 0.492, 0.641, 0.822, 0.894, and 0.917 from a requested 10^{-4} to 0.10 while the held-out null rate stays at or below the cutoff throughout.

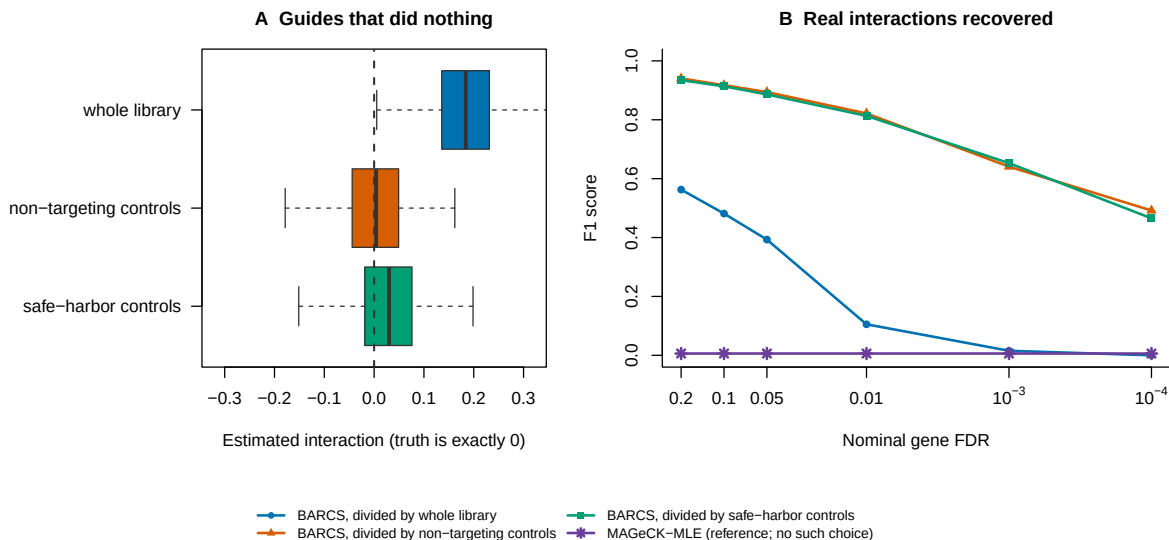


Figure 3: Denominator choice controls interaction recovery. (A) Estimated interactions among true-zero guides. A full-library denominator induces a positive shift because widespread depletion changes library composition; either control denominator removes it. (B) F1 across requested FDR thresholds. Non-targeting and safe-harbor denominators retain substantially more power than the full-library denominator. MAGECK-MLE is shown only as a reference because the simulated estimand is guide-level.

This design lies outside the intended scope of MAGECK-MLE. Its ranking performance is moderate, at AUROC 0.854, and its effect correlation is the highest observed, at 0.698, but it returns 1.7 calls from approximately 500 substantial interactions and its F1 is 0.006 at every cutoff from 10^{-4} to 0.20. This reflects a mismatch of estimand rather than a defect exposed by the simulation: the simulated truth is guide-level, so each sgRNA constitutes its own gene, which leaves a gene-level model no guides across which to pool and restricts its permutation null to 2,000 draws. We report the result to delimit the applicable range rather than to rank the method, and it is the quantization limit documented at genome scale, encountered from the opposite direction. An estimand defined per guide requires a guide-level test.

3 Methods

3.1 From CB² to BARCS

BARCS is fundamentally a guide-level coefficient model. Its required inputs are a guide-by-library count matrix, the corresponding full-library totals, and a sample design matrix; its primary output is one coefficient or contrast test per guide. Gene aggregation is downstream and optional. In particular, neither Fisher nor Stouffer combination is required to fit or interpret the BARCS regression. Some benchmarks below use an explicitly labeled Stouffer summary to compare with deposited gene-level results, whereas BARCS-GC uses the direct shared-effect Wald construction in Equations (28)–(31).

CB² adapts the beta-binomial weighted test of Baggerly et al. [17] to CRISPR pooled screens [7]. Its central insight is worth preserving: sequencing depth measures a guide proportion precisely, but it does not turn reads into independent biological replicates. The original method applies this insight to a two-group contrast. BARCS retains the same depth-aware, extra-binomial variance principle and changes the mean model from two group means to a design matrix. “Retains” refers to

the stochastic hierarchy and the sample-level information ceiling; it does not mean that the default logit fit reproduces the original finite-sample statistic. The legacy representation and asymptotic relationship are established below.

Table 4: What changes—and what does not—from the original CB² method to BARCS.

Feature	■ Original CB ²	■ BARCS
Primary question	Difference between two groups	Trend, adjusted association, interaction, or contrast
Sample design	Reference and comparison labels	Arbitrary full-rank R model matrix
Mean model	Two unrelated group proportions	$\text{logit}(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$
Effect	Difference or fold change between groups	Conditional log-odds coefficient or linear contrast
Dispersion	Separate beta-binomial weighting within each group	One guide-specific ρ shared across the fitted design
Estimator	Weighted group proportions	Feasible logistic IRLS
Variance principle	Binomial sequencing plus between-library beta-binomial heterogeneity	
Reference df	Group-variance df formula	Residual $m - q$
Adjustment	Pairwise stratification or preprocessing	Batch, donor, dose, time, interactions, splines, and other covariates
Implementation	Existing CB ² workflow	Additive <code>bbreg()</code> , <code>bb_contrast()</code> , and <code>bb_screen()</code> APIs with RcppArmadillo kernels

The regression bridge was described by Baggerly et al. [18] for sequencing count data with multiple groups and continuous covariates. The method combines logistic regression with extra-binomial variation and retains a coefficient-based t test. Here we state that construction in CRISPR-screen notation and use the beta-binomial, library-size-dependent variance of Williams [19]. An identity-link regression of raw proportions would allow fitted values outside $(0, 1)$, while an unweighted correlation would ignore unequal library sizes. The logit and beta-binomial weights solve those two problems directly.

3.2 Model

For guide g in sequenced sample i , let K_{gi} be its read count, let N_i be the full mapped-guide total for that sample, and let $Y_{gi} = K_{gi}/N_i$ be the observed guide proportion. Sample-level quantities such as dose, time, treatment, donor, and batch are predictors even when they are described biologically as experimental “outcomes.”

The model is fitted separately for every guide, so the guide index is suppressed below. Conditional on a latent sample-specific abundance P_i ,

$$K_i \mid P_i \sim \text{Binomial}(N_i, P_i), \quad (2)$$

$$P_i \sim \text{Beta}(\mu_i \kappa, (1 - \mu_i) \kappa). \quad (3)$$

The mean is linked to a row vector of sample covariates $\mathbf{x}_i^\top$ by

$$\text{logit}(\mu_i) = \log \left(\frac{\mu_i}{1 - \mu_i} \right) = \mathbf{x}_i^\top \boldsymbol{\beta}. \quad (4)$$

The design can contain an intercept, a continuous phenotype, batch indicators, interactions, spline terms, and other prespecified adjustment variables.

It is useful to express the beta precision κ as an intraclass correlation

$$\rho = \frac{1}{\kappa + 1}, \quad 0 \leq \rho < 1. \quad (5)$$

Marginally,

$$E(K_i) = N_i \mu_i, \quad (6)$$

$$\text{Var}(K_i) = N_i \mu_i (1 - \mu_i) \{1 + (N_i - 1) \rho\}, \quad (7)$$

$$\text{Var}(Y_i) = \mu_i (1 - \mu_i) \left\{ \frac{1 - \rho}{N_i} + \rho \right\}. \quad (8)$$

Equation (8) cleanly separates sequencing uncertainty, which decreases with N_i , from between-sample heterogeneity, which does not vanish when sequencing depth becomes arbitrarily large.

3.2.1 Interpretation of the continuous coefficient

Suppose the model contains a continuous phenotype d_i with coefficient β_d . Then $\exp(\beta_d)$ is the multiplicative change in the odds of observing that guide for a one-unit increase in d_i , conditional on the other covariates. Standardizing d_i before fitting makes $\exp(\beta_d)$ the odds ratio per sample standard deviation and makes guide effects directly comparable. For rare guides, odds and proportions are similar, so β_d is also approximately a log abundance ratio per unit of d_i .

3.3 Estimation by feasible IRLS

Let X be the $m \times q$ full-rank design matrix. Given current values of β and ρ , define

$$\eta_i = \mathbf{x}_i^\top \beta, \quad \mu_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}, \quad (9)$$

$$z_i = \eta_i + \frac{Y_i - \mu_i}{\mu_i(1 - \mu_i)}, \quad w_i = \frac{N_i \mu_i (1 - \mu_i)}{1 + (N_i - 1) \rho}. \quad (10)$$

The coefficient update is the weighted least-squares step

$$\hat{\beta}_{\text{new}} = (X^\top W X)^{-1} X^\top W \mathbf{z}, \quad W = \text{diag}(w_1, \dots, w_m). \quad (11)$$

This is weighted least squares on the logistic working response, not on the raw proportions.

For a fixed fitted mean, ρ is estimated from the Pearson equation

$$\sum_{i=1}^m \frac{(K_i - N_i \hat{\mu}_i)^2}{N_i \hat{\mu}_i (1 - \hat{\mu}_i) \{1 + (N_i - 1) \hat{\rho}\}} = m - q. \quad (12)$$

If the left side at $\rho = 0$ is no larger than $m - q$, the estimate is truncated at zero rather than claiming negative overdispersion. Equations (10)–(12) are alternated to convergence.

The weight has an important limiting behavior:

$$\lim_{N_i \rightarrow \infty} w_i = \frac{\mu_i(1 - \mu_i)}{\rho} \quad \text{when } \rho > 0. \quad (13)$$

Therefore a deeply sequenced sample cannot acquire unlimited leverage. Once sequencing uncertainty is small, biological heterogeneity sets the information ceiling.

3.4 T-based inference

At convergence, the model-based covariance estimator is

$$\widehat{\text{Cov}}(\hat{\beta}) = (X^\top \widehat{W} X)^{-1}. \quad (14)$$

The Pearson equation scales the residual variation to approximately one. If the dispersion estimate reaches its numerical upper boundary, the implementation additionally multiplies Equation (14) by the remaining Pearson scale.

Equation (14) is a plug-in, model-based covariance: it conditions on the estimated $\hat{\rho}$ as though it were fixed. It does not propagate uncertainty from estimating a separate dispersion for every guide. Referring the coefficient to t_{m-q} gives a heavier-tailed small-sample reference than a normal approximation, but it is not a formal correction for dispersion-estimation uncertainty. With few independent libraries, calibration must therefore be checked by simulation, phenotype permutation, or prespecified negative controls.

For coefficient β_j , the statistic

$$t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{\sqrt{\widehat{\text{Cov}}(\hat{\beta})_{jj}}} \quad (15)$$

is compared with a Student t_{m-q} distribution. The degrees of freedom are driven by the number of independent sequenced samples, not by the total read count. This is the crucial reason that a library with millions of reads does not create artificial certainty.

More generally, for a prespecified contrast vector $\mathbf{c}$,

$$t_{\mathbf{c}} = \frac{\mathbf{c}^\top \hat{\beta} - \delta_0}{\sqrt{\mathbf{c}^\top \widehat{\text{Cov}}(\hat{\beta}) \mathbf{c}}} \quad (16)$$

uses the same $m - q$ degrees of freedom. This covers pairwise group contrasts, average slopes in interaction models, and other one-degree-of-freedom hypotheses. Guide-wise two-sided p -values are adjusted using the Benjamini–Hochberg procedure [20].

3.4.1 From guide statistics to gene statistics

BARCS fits the count model guide by guide. The primary HT-29, FACS, simulation, GSE70038, and A375 benchmark scripts then use an explicit directional Stouffer summary, denoted BARCS-ST below, to obtain one gene-level statistic. For the j th valid guide targeting gene g , the two-sided guide p -value is converted back to a signed standard-normal score,

$$z_{gj} = \text{sign}(\hat{\beta}_{gj}) \Phi^{-1} \left(1 - \frac{p_{gj}}{2} \right). \quad (17)$$

If m_g valid guides target the gene, their combined score and two-sided gene-level p -value are

$$Z_g = \frac{1}{\sqrt{m_g}} \sum_{j=1}^{m_g} z_{gj}, \quad p_g = 2\Phi(-|Z_g|). \quad (18)$$

The reported gene effect is the median of the guide coefficient estimates,

$$\hat{\beta}_g = \text{median}_{j=1, \dots, m_g} \hat{\beta}_{gj}, \quad (19)$$

and gene-level false-discovery rates are obtained by applying Benjamini–Hochberg to the collection of p_g values. Consequently, concordant guide directions reinforce one another, discordant directions cancel, and one extreme guide has limited influence on the reported effect.

This aggregation is deliberately transparent, but it is not a native hierarchical gene model. Equations (17)–(18) give equal weight to valid guides and use the independence reference $\sqrt{m_g}$; shared sequence artifacts or other correlation can therefore make the gene statistic anti-conservative. A production extension should estimate guide reliability and within-gene dependence or use a hierarchical effect model. In the biological head-to-head benchmarks, this Stouffer–median procedure is used only for BARCS (and for the deliberately naive binomial comparator). MAGeCK, Chronos, BAGEL2, Waterbear, and MAUDE retain their native or deposited gene-level summaries. The exploratory DeWeirdt false-discovery audit used Fisher aggregation as stated in its Results description and is not pooled with these primary benchmark statistics.

Exchangeable normal guide-beta statistic. BARCS-NORM implements a different and deliberately simple model. It does not transform guide p -values and does not use guide-specific standard errors. Instead, for the m_g valid guides targeting gene g , it assumes

$$\hat{\beta}_{gj} \stackrel{\text{iid}}{\sim} N(\mu_g, \sigma_g^2). \quad (20)$$

The gene effect and between-guide standard deviation are estimated directly:

$$\hat{\mu}_g = \frac{1}{m_g} \sum_{j=1}^{m_g} \hat{\beta}_{gj}, \quad (21)$$

$$s_g^2 = \frac{1}{m_g - 1} \sum_{j=1}^{m_g} (\hat{\beta}_{gj} - \hat{\mu}_g)^2. \quad (22)$$

Testing $H_0 : \mu_g = 0$ gives

$$T_g = \frac{\hat{\mu}_g}{s_g / \sqrt{m_g}} \sim t_{m_g-1} \quad \text{under Equation (20) and } H_0. \quad (23)$$

Thus normally distributed guide betas lead to a Student t reference when their unknown σ_g is estimated from the same guides. The software also reports $2\Phi(-|T_g|)$ as a plug-in normal sensitivity value, but it is not the primary p -value because it treats the estimated spread as known. Multiple guides support this guide-consistency calculation; they do not become additional biological screen replicates.

Random-effects and moderated random-effects statistics. BARCS-RE (the implementation label BARCS-partial) retains each guide-specific regression standard error and models

$$\hat{\beta}_{gj} \mid \beta_g, \tau_g^2 \sim N\left(\beta_g, \text{SE}(\hat{\beta}_{gj})^2 + \tau_g^2\right). \quad (24)$$

It estimates τ_g^2 by the nonnegative DerSimonian–Laird estimator [21] and then pools guide effects with weights

$$w_{gj} = \left\{ \text{SE}(\hat{\beta}_{gj})^2 + \hat{\tau}_g^2 \right\}^{-1}, \quad \hat{\beta}_g = \frac{\sum_j w_{gj} \hat{\beta}_{gj}}{\sum_j w_{gj}}. \quad (25)$$

BARCS-RE-EB (implementation label **BARCS-EB**) stabilizes the noisy gene-specific heterogeneity estimate by shrinking it toward the pooled screen-wide excess- Q estimate τ_0^2 :

$$\hat{\tau}_g^2 = \frac{(m_g - 1)\hat{\tau}_g^2 + d_0\tau_0^2}{(m_g - 1) + d_0}, \quad d_0 = 4. \quad (26)$$

Both random-effects statistics are calibrated against prespecified negative-control genes when at least ten are available and otherwise against a robust whole-screen empirical null. In contrast, BARCS-NORM obtains its spread and reference degrees of freedom entirely within each gene. These three alternatives and BARCS-ST reuse exactly the same fitted guide coefficients; only the guide-to-gene inference changes.

Guide-consistency statistic for the one-replicate boundary. When sample replication is insufficient, converting a guide statistic through a low-degree-of-freedom t distribution can erase useful ranking information. We therefore prespecified a separate exploratory gene statistic, BARCS-GC, for the $R = 1$ CRISPulator analysis. It uses the uncalibrated guide coefficient and model-based standard error to estimate one shared gene effect. With

$$w_{gj} = \text{SE}(\hat{\beta}_{gj})^{-2}, \quad (27)$$

the inverse-variance estimate, its standard error, and raw Wald statistic are

$$\hat{\beta}_g = \frac{\sum_{j=1}^{m_g} w_{gj} \hat{\beta}_{gj}}{\sum_{j=1}^{m_g} w_{gj}}, \quad \text{SE}(\hat{\beta}_g) = \left(\sum_{j=1}^{m_g} w_{gj} \right)^{-1/2}, \quad T_g = \frac{\hat{\beta}_g}{\text{SE}(\hat{\beta}_g)}. \quad (28)$$

This is a direct fixed-effect estimate of the common guide coefficient, not a Fisher or Stouffer combination of guide p -values. Its additional assumption is that the guide coefficients target one common gene effect; the reported direction-agreement fraction exposes gross heterogeneity.

Because T_g does not have a trustworthy sample-replicate reference distribution at $R = 1$, it is calibrated across genes. Let c_0 be the all-gene median and

$$s_{\text{MAD}} = \frac{\text{median}_g |T_g - c_0|}{\Phi^{-1}(0.75)}. \quad (29)$$

When at least ten valid control genes are present, let c_C be their median and let

$$s_C = \frac{Q_{0.95}\{|T_g - c_C| : g \in C\}}{\Phi^{-1}(0.975)}. \quad (30)$$

The calibrated statistic and two-sided empirical-null p -value are

$$Z_g^{\text{GC}} = \frac{T_g - c_C}{\max(1, s_{\text{MAD}}, s_C)}, \quad p_g^{\text{GC}} = 2\Phi(-|Z_g^{\text{GC}}|), \quad (31)$$

followed by Benjamini–Hochberg adjustment. If too few control genes are available, c_0 replaces c_C and s_C is omitted. At least three finite guide coefficient estimates are required per gene. The all-gene MAD assumes that fewer than half of tested genes are active.

BARCS-GC measures reproducibility across independently designed perturbations; it does not turn guides into independent biological samples. Its p -value is consequently labeled empirical-null guide-consistency evidence, not sample-level confirmation. The ordinary BARCS result remains the primary analysis when biological replicates are available, and independent experimental replication remains necessary for confirmatory claims.

3.4.2 Longitudinal HT-29 benchmark

Guide counts for the HT-29 time course of Tzelepis et al. [12] were obtained from Data S2 of that article, file `grna_counts_HT29C-clone3_d7-25-HT1080.txt`, retrieved from the Europe PMC supplementary bundle for PMC5081405. The three sequencing columns reported at each of days 7, 10, 13, 16, 19, 22, and 25 were summed to one column per harvest and combined with the pDNA measurement, and guides with fewer than 30 pDNA reads were removed, leaving 86,882 guides targeting 17,995 genes. Library totals were computed before that filter and never recomputed, so the beta-binomial denominator remains the full mapped-read total for each sample; the filter removes 4.2% of guides but only 0.10% of reads, uniformly across samples. BARCS and official MAGeCK-MLE 0.5.9.5 were both fitted to this eight-column matrix under an identical design, in which pDNA is day 0 and the tested predictor is day divided by 25, so the two continuous-time fits differ only in the likelihood. A minimum total count of 30 was applied within the BARCS fit.

Comparator effects were not recomputed. The joint Chronos, per-endpoint MAGeCK, and per-endpoint BAGEL2 effect matrices were taken as deposited in Chronos Figshare article 14067047 [13], as were the reference essential and nonessential gene lists; the corresponding individual file identifiers are 26548532, 26548541, 30847513, 26548550, and 26548553. As Those two matrices carry one column per day; the day-25 column is used directly, so the comparator is a single observed endpoint at the deepest time point rather than a summary across days.

Two external annotations define the evaluation sets. Positives are the deposited reference essential genes. Negatives are HT-29 genes with DepMap Public 20Q2 expression below 0.5, taken from the ACH-000552 column of Figshare file 26548487, which is the expression source used for the same threshold in the deposited Chronos analysis. The copy-number audit uses the ACH-000552 column of Figshare file 26548484 [22], and correction was performed with MAGeCK 0.5.9.5’s official `piecewise-cnv-norm` normalizer applied after model fitting. Because that profile was measured independently of the screen, cell-line drift is a limitation of the copy-number analysis.

The analysis is reproduced by `examples/chronos_tzelepis_benchmark.R`, which calls the MAGeCK executable through `scripts/mageck_compat.py` and `scripts/mageck_cnv_correct.py` and writes the tables reported here. Provenance for each downloaded input is recorded in `data/raw/chronos/README.md`.

3.4.3 CRISPulator FACS benchmark

We used CRISPulator 0.5.1’s documented interfaces for library construction, transfection, FACS selection, and sequencing [14]. The committed Julia environment pins the tagged source revision. For each simulation seed, one CRISPRn guide library was constructed and reused across four independently seeded transfection, sorting, and sequencing replicates. The default CRISPulator phenotype mixture was retained: 75% inactive, 5% negative-control, 10% phenotype-increasing, and 10% phenotype-decreasing genes. The headline screens contained 10,000 genes and five guides per gene, so 50,000 guides, which is the order of a genome-wide library rather than a pilot. The default guide-quality mixture—90% complete-knockout guides and 10% low-activity guides—was retained. Transfection representation was 500 cells per guide, Gaussian phenotype noise was 2, sorting representation was 50 cells per guide, and sequencing depth was 50 reads per guide per sample. Multiplicity of infection was 0.20 for the main analysis and 0.30 for the supplementary analysis; 0.20 is the point at which the single-integration approximation these guide-level models share is most defensible, and 0.30 stresses it. Both multiplicity of infection and the high-quality-guide fraction are exposed as simulation parameters, as is the number of genes and the number of independent screen replicates. The three genome-scale benchmark seeds were 20250724 through 20250726; the

supporting 400-gene analyses below use 20250724 through 20250728.

Two BARCS gene statistics are carried through the genome-scale benchmark. BARCS-ST is the historical calibrated signed- z statistic. BARCS-MOD reuses the identical guide fits after guide-dispersion moderation, so the two differ only in how the guide variance is estimated, not in the gene combiner. The exchangeable-normal, random-effects, and moderated random-effects statistics are retained for the real-data comparisons but are not carried through the genome-scale simulation, where the question is whether moderation changes the calibration–power tradeoff rather than which of several pooling rules is best. Official MAGeCK-MLE 0.5.9.5 is included, with its ordered marker affinely mapped onto zero–one so that the low samples of the reference replicate form the all-zero reference row the MAGeCK initializer requires. Because MAGeCK-MLE omits negative-control genes from its gene summary once their sgRNAs are declared, all methods in the genome-scale benchmark are scored on the genes every method returned a finite result for, 9,475–9,517 of 10,000 per run. The negative-control diagnostic is instead taken from each method’s own output, and is therefore unavailable for MAGeCK-MLE.

We additionally performed a one-factor-at-a-time sensitivity analysis around the baseline $(\text{MOI}, q, G, R) = (0.25, 0.90, 400, 4)$, where q is the high-quality-guide fraction, G is the number of genes, and R is the number of independent screen replicates. The evaluated levels were $\text{MOI} \in \{0.10, 0.25, 0.40\}$, $q \in \{0.60, 0.75, 0.90\}$, $G \in \{100, 400, 1000\}$, and $R \in \{1, 3, 4, 6\}$. After removing duplicate baseline settings, this yielded ten scenarios; each used the same five prespecified seeds, for 50 simulations. One parameter changed at a time, so its main sensitivity remained interpretable. For each seed and scenario, we paired BARCS and MAGeCK-MLE results from the low–bulk–high design and reported differences as BARCS minus MAGeCK-MLE. The executable benchmark also supports the complete $3 \times 3 \times 3 \times 4$ factorial (108 scenarios and 540 simulations across five seeds) when interaction effects are the target.

The $R = 1$ setting was prespecified as a diagnostic boundary rather than a confirmatory design. Its three low–bulk–high samples and two regression coefficients leave only one residual degree of freedom. The corresponding two-tail fit has two samples and two coefficients, leaves zero residual degrees of freedom, and was therefore not fitted. All pairwise-design comparisons and the primary performance interpretation use $R \geq 3$; the one-replicate result is shown separately to expose the consequence of insufficient replication. BARCS-GC was evaluated only at $R = 1$ using Equations (28)–(31); it was not used to replace ordinary BARCS in supported replicated designs.

CRISPulator’s arbitrary percentile ranges were set to $(0, 0.25)$ for low, $(0.75, 1)$ for high, and $(0, 1)$ for bulk. Thus bulk deliberately overlaps the tails and is not a third component of a multinomial partition. An input sample was generated as an independent sequencing draw from the same guide frequency vector as bulk. Low and high received phenotype scores equal to the means of their corresponding standard-normal intervals,

$$d_{\text{low}} = -1.271, \quad d_{\text{high}} = 1.271,$$

while bulk received $d_{\text{bulk}} = 0$. BARCS and MAGeCK-MLE fitted this score with replicate indicators [6]. Tail-only fits removed bulk without changing the low and high counts. BARCS guide tests were scaled with the simulated negative-control guides before directional Stouffer aggregation; MAGeCK-MLE used its native gene-level Wald result. The comparators are CRISPR-specific gene callers rather than general count regressions, because that is the choice a screen analyst faces. CRISPhieRmix [15] was run on guide-level $\log_2$ fold changes from a DESeq2 Wald fit of the same design [2], which is the input its documentation specifies; DESeq2 therefore appears as a preprocessing step and is not itself scored. CRISPhieRmix was run with its bimodal alternative, because this screen contains both phenotype-increasing and phenotype-decreasing genes and the one-sided default would score

only one direction. Its gene effect is the mean guide $\log_2$ fold change and its local false-discovery rate stands in for a p -value wherever a ranking is required, since the model reports neither a p -value nor a signed gene effect. Reported runtimes cover the primary low-bulk-high model fit and exclude simulation, file input/output, and guide-to-gene aggregation.

Genes from the increasing and decreasing classes were positives. Ranking used the two-sided gene significance score. AUROC and average precision measure active-gene discrimination. Effect recovery is Spearman correlation between the inferred gene effect and CRISPulator’s median theoretical guide phenotype among active genes. Directional recall is the fraction of active genes with gene FDR below 0.10 and an effect sign matching the simulated class. Realized FDP is the fraction of called genes that were inactive, and F1 uses active versus inactive status at the same FDR threshold.

3.4.4 simCRISPR interaction benchmark

CRISPulator assigns each guide a single effect and sorts cells on a phenotype, so it cannot generate the design BARCS is aimed at. simCRISPR [16] simulates a factorial screen instead: knockout induction crossed with treatment, propagated over several days of cell growth, with PCR amplification and sequencing applied afterwards. Every sgRNA therefore carries a knockout effect, a treatment effect, and an interaction between the two, and that interaction is a coefficient in a design matrix rather than a contrast of two arms.

Each of three seeds generated 2,000 sgRNAs, of which 300 were non-targeting and 200 were safe-harbor controls, in three independent replicates of each of the four arms, so twelve libraries and eight residual degrees of freedom. Growth was logistic. The alternative exponential model is unbounded: over five days a single guide reached a fifth of the library and library sizes spanned a 175-fold range, which does not represent a pooled screen. Initial library columns were dropped, since they are not part of the factorial contrast. Amplification used the package defaults and each library was sequenced to 500 reads per guide. BARCS fitted $\text{logit}(\mu_{gi}) = \beta_0 + \beta_1 k_i + \beta_2 t_i + \beta_3 k_i t_i$ for knockout indicator k_i and treatment indicator t_i , and tested β_3 .

Three denominators were compared. The library denominator is the full column total, BARCS’s default. The two control denominators hold a chosen control class at a constant share, rescaled to the mean library size and rounded, so that they remain interpretable as library sizes. The two control classes are not interchangeable: safe-harbor guides are cut and carry the DNA-damage response, non-targeting guides are neither. Each control class was split in half, one half normalizing and calibrating and the other held out, so that no guide used to fit the null was also scored against it. MAGeCK-MLE received the same design matrix with an explicit interaction column, the same calibration-half controls, and one sgRNA per gene, because the simulated truth is guide-level.

CRISPhieRmix was not run on these data. It borrows strength across the guides within a gene, whereas simCRISPR assigns every sgRNA an independent interaction, so grouping guides into synthetic genes would impose the structure that the model exploits.

Every targeting guide receives a nonzero interaction, most far too small to resolve, so a magnitude was fixed before scoring: guides with $|\text{INT}| \geq 0.2$ are the positives and the held-out non-targeting guides, whose interaction is exactly zero, are the negatives. Targeting guides below that magnitude are scored for effect recovery but excluded from the detection metrics. Effect recovery is the Spearman correlation between the estimated interaction coefficient and the simulated one across all targeting guides, which needs no threshold. The held-out control call rate and the safe-harbor call rate are reported separately: the first measures false positives against a ground-truth zero, the second measures how often the cutting response alone is mistaken for an interaction.

3.4.5 Liang Cas13 longitudinal fitness benchmark

We analyzed the transcriptome-scale RfxCas13d fitness screens of Liang et al. [10] in HAP1, HEK293FT, K562, MDA-MB-231, and THP1 cells. The library contains 56,322 guides targeting 5,496 lncRNAs, 3,156 protein-coding genes, and 1,000 non-targeting controls. Deposited Table S2 contains guide measurements at days 0, 7, and 14. The previous endpoint analysis selected only days 0 and 14; the present analysis uses all three time points.

The deposited values are fractional after median-of-ratios normalization, ComBat correction, and replicate-outlier processing. They are therefore not literal sequencing counts. BARCS requires integer observations, so each value was rounded once to the nearest pseudo-count and the identical rounded matrix was supplied to every newly fitted method. The mean absolute rounding change was approximately 0.25 and the maximum was less than 0.5 in every cell line. Consequently, this benchmark is a same-input sensitivity analysis rather than a likelihood-faithful raw-count comparison.

BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma-voom fitted a linear longitudinal effect with time scaled from 0 to 1 over 14 days. A replicate indicator was included when both replicate trajectories contained all three time points. K562 lacks one deposited day-0 replicate; all methods therefore used an unblocked time slope in that cell line rather than fitting an incomplete matched trajectory. BARCS guide statistics were calibrated with the non-targeting controls and combined with the prespecified directional Stouffer summary. BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma-voom used two-sided guide or Wald probabilities. For the four guide-level regression methods, coefficient direction was retained in the signed Stouffer score; the resulting gene probability was two-sided and was adjusted by the Benjamini–Hochberg procedure. MAGeCK-RRA’s negative-selection probability and the deposited Liang RRA probability were retained in their native depletion-tail form.

Known-essential protein-coding genes were positives. Cell-line-specific nulls were targeted lncRNAs with TPM equal to zero in both available expression assays. We reported average precision, essential-control recall at an empirical 5% null false-positive rate, the absolute deviation of the null $p < 0.05$ fraction from 0.05, and directional essential-control recall at gene FDR 0.10. The repository also provides a restartable raw-read pipeline for BioProject PRJNA1344834 using the stated Cutadapt anchors [23] and Bowtie -v 1 -m 3 -best -q [24].

3.4.6 Legacy representation and local-equivalence result

We now separate an algebraic compatibility result from the substantive estimating-equation connection. The first result reproduces CB² after its weighted group summaries have already been computed; because a saturated two-cell GLS can represent any pair of independent group summaries, this result should not be interpreted as proof that the implemented `bbreg()` estimator strictly nests the original estimator. For the original test, let

$$\hat{p}_A = \sum_{i \in A} a_i Y_i, \quad \hat{p}_B = \sum_{i \in B} b_i Y_i, \quad \sum_{i \in A} a_i = \sum_{i \in B} b_i = 1,$$

where the depth- and dispersion-dependent weights a_i, b_i are those estimated by CB². Let its estimated group variances be $V_A > 0$ and $V_B > 0$. The original guide statistic and reference degrees of freedom are

$$T_{\text{CB2}} = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{V_A + V_B}}, \tag{32}$$

$$\nu_{\text{CB2}} = \frac{(V_A + V_B)^2}{V_A^2/(n_A - 1) + V_B^2/(n_B - 1)}. \tag{33}$$

Lemma 1 (Legacy CB² representation). *Let X have row $(1, 0)$ for samples in group A and row $(1, 1)$ for samples in group B . Define the working precision matrix*

$$\Omega = \text{diag} \left(\left\{ \frac{a_i}{V_A} : i \in A \right\}, \left\{ \frac{b_i}{V_B} : i \in B \right\} \right).$$

Then the identity-link generalized least-squares estimator satisfies

$$\hat{\beta}_{\text{GLS}} = (X^\top \Omega X)^{-1} X^\top \Omega \mathbf{Y} = \begin{pmatrix} \hat{p}_A \\ \hat{p}_B - \hat{p}_A \end{pmatrix},$$

and

$$(X^\top \Omega X)^{-1} = \begin{pmatrix} V_A & -V_A \\ -V_A & V_A + V_B \end{pmatrix}.$$

Consequently, the GLS t statistic for the slope is exactly T_{CB2} .

Proof. Because the weights within each group sum to one, their groupwise total working precisions are

$$\sum_{i \in A} \frac{a_i}{V_A} = \frac{1}{V_A}, \quad \sum_{i \in B} \frac{b_i}{V_B} = \frac{1}{V_B}.$$

Hence

$$X^\top \Omega X = \begin{pmatrix} V_A^{-1} + V_B^{-1} & V_B^{-1} \\ V_B^{-1} & V_B^{-1} \end{pmatrix}, \quad X^\top \Omega \mathbf{Y} = \begin{pmatrix} \hat{p}_A/V_A + \hat{p}_B/V_B \\ \hat{p}_B/V_B \end{pmatrix}.$$

Direct inversion gives the displayed covariance matrix and multiplication gives the displayed coefficient vector. Its slope standard error is $\sqrt{V_A + V_B}$, proving the result. $\square$

Corollary 1 (Legacy-compatibility identity). *If the slope statistic in Lemma 1 is referred to $t_{\nu_{\text{CB2}}}$ using Equation (33), its two-sided p -value is identical to the original CB² p -value. Thus identity-link GLS algebra reproduces the original two-group inference on its group-summary scale. This corollary is a compatibility check, not a claim of finite-sample equality with the default logit `bbreg()` fit.*

The default implementation uses the logit link because it respects the parameter space for arbitrary designs. It is therefore important to prove a second, weaker relationship rather than claim finite-sample identity.

Proposition 1 (First-order equivalence of the logit contrast). *Suppose the two groups are independent, their weighted proportion estimators are consistent, their groupwise and pooled precision estimators converge to a common κ , and under the null or a sequence of local alternatives $p_A, p_B \rightarrow p \in (0, 1)$ with $\hat{p}_B - \hat{p}_A = O_p(n^{-1/2})$. Let $h(p) = \text{logit}(p)$ and define the delta-method statistic*

$$T_{\text{logit}} = \frac{h(\hat{p}_B) - h(\hat{p}_A)}{\sqrt{h'(\hat{p}_A)^2 V_A + h'(\hat{p}_B)^2 V_B}}.$$

Then

$$T_{\text{logit}} - T_{\text{CB2}} \xrightarrow{p} 0.$$

Under a correctly specified common-dispersion beta-binomial model, the binary-design IRLS Wald statistic has the same first-order expansion. If $n_A, n_B \rightarrow \infty$ through increasing numbers of independent biological libraries, both its residual degrees of freedom and Equation (33) diverge, so their reference distributions also converge to the same standard normal limit. Holding n_A, n_B fixed while only increasing read depths N_i is not this asymptotic regime.

Proof. The mean-value theorem and consistency give

$$h(\hat{p}_B) - h(\hat{p}_A) = h'(p)(\hat{p}_B - \hat{p}_A) + o_p(n^{-1/2}),$$

where $h'(p) = \{p(1-p)\}^{-1}$. Continuity of h' gives

$$\sqrt{h'(\hat{p}_A)^2 V_A + h'(\hat{p}_B)^2 V_B} = h'(p) \sqrt{V_A + V_B} \{1 + o_p(1)\}.$$

The common positive factor $h'(p)$ cancels, yielding $T_{\text{logit}} = T_{\text{CB2}} + o_p(1)$. It remains to connect this transformed cell-mean statistic to IRLS. Write $\rho = 1/(\kappa + 1)$. Within a group having mean μ_g , the beta-binomial quasi-score for its logit cell mean is proportional to

$$\sum_{i \in g} \frac{N_i}{\kappa + N_i} (Y_i - \mu_g).$$

The original CB² weight is

$$\left(\frac{1}{\kappa} + \frac{1}{N_i} \right)^{-1} = \frac{\kappa N_i}{\kappa + N_i}.$$

The factor κ disappears on normalization, so for fixed common dispersion the two methods use identical within-group relative weights. Under the stated common-dispersion model, the separately estimated legacy precisions and the IRLS precision converge to the same κ ; the resulting binary-design cell means therefore have the same first-order expansion. Finally, both t distributions approach the standard normal as their degrees of freedom diverge. $\square$

These results establish two deliberately separate claims. The original statistic has an *algebraically exact legacy representation* after its group summaries are computed, while the implemented logit statistic is *locally asymptotically equivalent* in the unadjusted binary design under the stated common-dispersion assumptions. Strict finite-sample nesting of the implemented estimator is not claimed. In finite samples, `bbreg(~group)` need not equal `measure_sgrna_stats()`: the effect scale, dispersion pooling, working weights, and degrees of freedom differ. What regression adds is an estimable coefficient for continuous predictors, adjustment variables, and interactions; what it changes must be assessed by calibration and benchmark data rather than hidden behind the word “generalization.” The executable example `examples/cb2_generalization_proof.R` verifies Lemma 1 to machine precision and illustrates the convergence in Proposition 1. In its deterministic guide, original CB² and the GLS contrast both give effect 9.059929×10^{-4} , standard error 1.113105×10^{-4} , $t = 8.139330$, $\nu = 5.647255$, and two-sided $p = 2.512928 \times 10^{-4}$; the maximum numerical discrepancy is 1.78×10^{-15} . In the link-scale illustration, shrinking the two-group proportion difference from 8×10^{-4} to 2.5×10^{-5} reduces the absolute gap between raw and logit statistics from 0.0583 to 0.000151.

4 Software and usage

BARCS is distributed as an R package whose regression layer runs on base R alone, so it remains usable without a compiler. Its two inner kernels, the weighted IRLS cross-products and the symmetric solve, are implemented in RcppArmadillo [25] within the package itself and are selected automatically when the compiled library is available; the base R path is retained as an exactly equivalent fallback. The package imports CB² rather than vendoring it, so the original two-group functions and the FASTQ quantifier remain available and unchanged from their own package: BARCS is an additive analysis path rather than a breaking replacement.

```

library(BARCS)

# One guide: K contains guide counts; N contains sample library sizes.
fit <- bbreg(
  count    = K,
  total    = N,
  formula  = ~ scale(dose) + batch,
  data     = sample_data
)
summary(fit)

# A named contrast; unspecified coefficient weights are zero.
bb_contrast(fit, c("scale(dose)" = 1))

# A complete guide-by-sample screen.
result <- bb_screen(
  counts   = count_matrix,
  totals   = N,
  data     = sample_data,
  formula  = ~ scale(dose) + batch,
  term     = "scale(dose)",
  gene     = guide_annotation$gene,
  ncores   = 4
)

# Optional empirical-null scaling when prespecified negative controls exist.
result <- bb_calibrate_controls(
  result,
  control  = guide_annotation$gene == "Non-Targeting Control"
)
result[order(result$fdr), ]

# Optional direct shared-effect gene inference; no Fisher/Stouffer step.
gene_result <- bb_gene_consistency(
  result,
  control  = guide_annotation$gene == "Non-Targeting Control"
)

```

Column order in the count matrix must match row order in `sample_data`. The phenotype should be defined before looking at guide results, and all independent biological replicates should remain separate. For the negative-binomial benchmark, the project calls the official `mageck mle` executable with the same design matrix rather than embedding a partial MAGeCK reimplementaion. On the development machine, the installed package analyzes 10,000 guides in 4.73 seconds serially and 1.36 seconds with four forked workers. The longitudinal analysis fits 86,882 filtered guides across eight aggregated time points in 21.4 seconds with four workers (4,068 guides per second); 86,840 fits reached the strict convergence tolerance, and all fits returned finite time coefficients.

5 Discussion

BARCS extends the beta-binomial sampling principle established by CB². Pooled-guide counts are represented as proportions conditional on observed library totals, and the mean is generalized from two groups to an arbitrary design matrix. The resulting estimand is an ordinary coefficient: a time slope, dose response, adjusted treatment effect, interaction, or prespecified contrast.

Goodness-of-fit comparisons in the original CB² study favored the beta-binomial representation of pooled-guide counts [7]. The present results extend that advantage to regression designs. BARCS maintained comparable essential-gene discrimination in HT-29 and substantially improved longitudinal null calibration in the Liang screens while using the known library denominator directly.

The two longitudinal analyses are the principal evidence for generalization. HT-29 demonstrates that BARCS can analyze eight time points without reducing them to an endpoint contrast. The Liang reanalysis uses days 0, 7, and 14 jointly across five cell lines, rather than the previous day-0/day-14 comparison. Together they show that the method performs the analysis that CB² cannot express. The IL2RA experiment extends the same argument to an ordered phenotype with donor adjustment, where intermediate bins improved validated-regulator recovery over an outer-bin comparison.

The simulations refine the performance claim. BARCS-MOD provided the best F1 at matched realized false-discovery proportions across the evaluated operating points. The factorial simulation further showed that replacing the full library with prespecified control guides increased interaction F1 from 0.481 to 0.917. The sampling model, normalization contract, dispersion estimator, and calibration procedure therefore form one analysis pipeline.

The external false-discovery audit reinforces the importance of that contract (Supplementary Table S4). Preserving immutable full-library totals reduced the Avana null-only result from 152 reported false discoveries to zero on the identical evaluation genes. In the independent PSN1 null split, BARCS produced 4.81% guide $p < 0.05$ and no gene discoveries. These results show that the denominator must remain part of the beta-binomial sampling model after guide filtering.

The appropriate method remains design dependent. CB² is a direct choice for one prespecified two-group contrast. BARCS is appropriate when independent libraries lie on a quantitative or multivariable design. Waterbear is preferable when joint FACS-bin dependence is central, and Chronos is preferable when knockout timing and population dynamics are the scientific model. Negative controls should be used for calibration when available, and post-fit copy-number correction should be accepted only after a null-gene diagnostic. These are not exceptions to the main claim; they define its domain.

6 Limitations

1. **Independent samples.** The t_{m-q} reference distribution assumes independent sequenced libraries. Repeated measures, matched samples, or multiple libraries from one biological unit require a subject-level correlation model or a blocked permutation. FACS bins from one donor are a compositional partition, not four independent biological replicates; the GSE242880 control calibration is a diagnostic sensitivity analysis, whereas a purpose-built joint model such as Waterbear represents that structure directly. Likewise, a 0–100% bulk pool overlaps its sorted tails. The CRISPulator analysis includes that requested design to measure its empirical behavior, not to redefine bulk as an independent third bin.
2. **Enough sample-level degrees of freedom.** Read depth cannot replace biological replication. With very few samples, guide-wise $\hat{\rho}$ is noisy and coefficient tests may be poorly calibrated. The model-based covariance treats $\hat{\rho}$ as a fixed plug-in value, and the t_{m-q} reference does not

formally propagate its estimation uncertainty. The local-equivalence result likewise requires increasing numbers of independent libraries; deeper sequencing at fixed replication is not sufficient. The optional `bb_calibrate_controls()` procedure additionally requires all control guides to share the same residual degrees of freedom, as they do when every guide uses the same complete design. Guides fitted with heterogeneous designs require stratified or redesigned calibration. The optional `bb_gene_consistency()` statistic can recover concordant multi-guide signal when sample-level degrees of freedom are inadequate, but its empirical-null p -values measure guide consistency. They do not repair the missing biological replication and should not be described as confirmatory sample-level inference. Dispersion shrinkage across guides or permutation of the phenotype should be evaluated before treating this prototype as a production method.

3. **Mean-model form.** A single slope assumes a linear relationship on the logit scale. Residual plots should be checked. A spline or a prespecified nonlinear term is preferable when the dose response is curved. This matters directly in viability time courses: incomplete knockout can produce finite depletion saturation, which Chronos models mechanistically but a single beta-binomial slope does not.
4. **Sparse and separated guides.** Guides with all zero counts or with counts confined to one end of the phenotype range can cause boundary fits. Low-count filtering should be chosen without reference to the tested phenotype. A penalized or likelihood-ratio procedure is a useful future extension for severe separation.
5. **Compositional counts.** Guide counts share a fixed library total. The marginal beta-binomial model handles depth and guide-specific overdispersion but does not remove global composition shifts. Stable nontargeting controls and sensitivity analyses with alternative size factors are advisable. If guides are filtered for testing or benchmarking, the `totals` argument must retain the unfiltered mapped-read totals. Recomputing totals after row subsetting changes the likelihood and can materially change calibration.
6. **Guide-to-gene inference.** The BARCS regression itself is guide-level and requires no Fisher or Stouffer combination. Some replicated benchmark analyses use a directional Stouffer summary only as a transparent, explicitly labeled comparison adapter. BARCS-GC instead estimates one shared guide coefficient by inverse-variance weighting and calibrates its Wald statistic against a robust gene-level empirical null for the one-replicate diagnostic. That fixed-effect construction assumes a common gene effect, while its all-gene MAD assumes that fewer than half of genes are active. The direction-agreement output should be inspected for guide heterogeneity. We additionally evaluated BARCS-NORM, which treats the fitted guide coefficients within a gene as exchangeable draws $\hat{\beta}_{gj} \sim N(\mu_g, \sigma_g^2)$ and tests the sample mean with a t_{m_g-1} reference because σ_g is estimated. This literal normal model was calibrated but underpowered with three to six guides: across the 50 supporting 400-gene CRISPulator runs its average precision was 0.706 and its directional recall at FDR 0.10 was 0.159. Replacing the t reference by a plug-in normal tail would incorrectly treat the estimated σ_g as known. A production hierarchical gene test should therefore borrow variance information across genes and be benchmarked separately, ideally with held-out controls or phenotype permutations that preserve cross-guide dependence.
7. **Processed longitudinal inputs.** The Liang deposited values are normalized, batch-corrected pseudo-counts, so the fitted likelihood does not retain a literal raw-count interpretation. The three-time-point analysis is longitudinal, but raw-read confirmation remains necessary before

using it as a distributional validation. Likewise, post-fit CNV correction should be accepted only after a null-gene diagnostic: it succeeds in A375 and over-corrects the longitudinal HT-29 screen.

7 Conclusion

BARCS generalizes the depth-aware beta-binomial principle of CB^2 from a two-group comparison to longitudinal, covariate-adjusted, and factorial pooled-screen designs. It conditions each guide count on the unfiltered library total, assigns inferential degrees of freedom to independent libraries, and reports a prespecified coefficient or contrast.

The evidence supports a focused conclusion. BARCS achieved parity with specialist longitudinal methods in HT-29, used all three time points in the Liang Cas13 screens, improved validated-regulator recovery in a donor-adjusted ordered-bin analysis, and led F1 at matched realized error rates in genome-scale simulation. These results establish the regression generalization and its calibrated performance across multiple screen designs.

The remaining priorities are a hierarchical gene model, correlation-aware inference for repeated biological units and compositional bins, nonlinear time and dose effects, and raw-count confirmation of the Liang analysis. Within its stated domain, BARCS supplies the missing link between CB^2 's beta-binomial sampling model and modern multivariable screen designs.

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Supplementary information

Supplementary material referenced from the main text.

Supplementary Table S1: Operational definition of the problem BARCS addresses. Referenced from the Introduction.

Component	Definition
Observed data	Guide counts K_{gi} , unfiltered full-library totals N_i , and a prespecified sample design matrix X
Primary estimand	A guide-level coefficient or contrast $\mathbf{c}^\top \boldsymbol{\beta}_g$ conditional on the other columns of X
Information unit	An independent biological screen unit represented by a sequenced library; additional reads refine that unit but do not create another one
Primary output	Guide effect, standard error, t statistic, residual degrees of freedom, p -value, FDR, and dispersion diagnostics
Optional gene output	An inverse-variance shared-effect Wald statistic with empirical-null calibration; no Fisher or Stouffer step is required
Not identified	Unobserved single-cell phenotypes, dependence among bins from one biological pool, mechanistic growth dynamics, or uncertainty that would have required missing biological replicates

7.1 Cross-benchmark calibration diagnostics

Prespecified negative controls provide an empirical check on reported p -values. Under a calibrated null, approximately 5% of non-targeting guides, safe-harbor targets, simulated null genes, or non-expressed transcripts should satisfy $p < 0.05$. This diagnostic complements ranking and recall because a method can gain apparent sensitivity by returning anti-conservative probabilities.

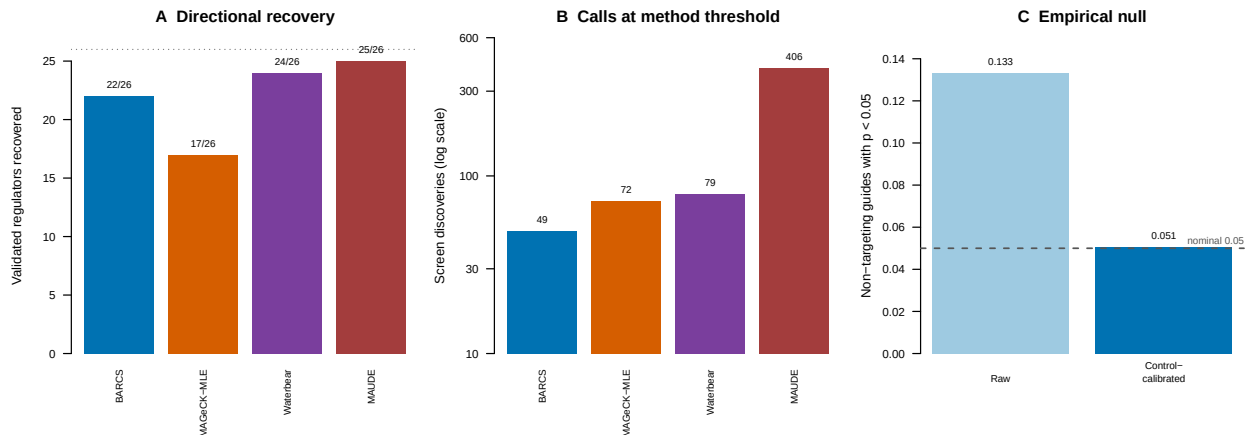
The clearest experimental example was the IL2RA FACS screen. The uncalibrated BARCS fit yielded $p < 0.05$ for 0.133 of non-targeting guides; negative-control calibration reduced that fraction to 0.051 while retaining 22 of 26 validated regulators. Calibration is therefore not an automatic consequence of the estimator: it depends on the complete analysis contract, including the denominator, design, and use of controls.

The longitudinal Liang analysis led to the same distinction. Across five cell lines, BARCS had a macro-averaged null $p < 0.05$ rate of 0.076 and hence an absolute calibration error of 0.035. The matched longitudinal MAGeCK-MLE analysis had a null rate of 0.153 and an error of 0.103. General RNA-seq count models achieved higher ranking sensitivity but had null rates between 0.139 and 0.153.

These results argue for reporting empirical null behavior alongside power, not for selecting one method in every design. BARCS was comparatively well calibrated among procedures estimating the same longitudinal slope.

7.2 Copy-number sensitivity in the longitudinal HT-29 screen

We extracted the HT-29 profile (ACH-000552) from DepMap Public 20Q2 [22] and repeated the same copy-number audit used for A375. Before correction, effect-copy-number Spearman correlation among unexpressed genes is already small: -0.072 for BARCS, -0.075 for official continuous-time MAGeCK, -0.048 for deposited Chronos joint effects, -0.057 for the published day-25 MAGeCK effects, and $+0.029$ for the published day-25 BAGEL2 effects. Applying MAGeCK 0.5.9.5's official



Supplementary Figure S1: Supporting ordered-bin IL2RA benchmark. (A) Directional recovery of 26 independently validated regulators. (B) Total discoveries at each method’s threshold (log scale). (C) Empirical non-targeting-guide $p < 0.05$ rate before and after BARCS control calibration; the dashed line is the nominal rate.

piecewise normalizer makes the association larger in absolute value rather than smaller: -0.145 for BARCS and -0.149 for MAGECK. Thus the improvement observed in A375 does not generalize automatically.

The piecewise correction estimates one global effect–CNV trend across genes in one cell line and applies it after model fitting. When the nuisance association is weak and biological dependencies are not exchangeable across copy-number regions, the fitted trend can over-correct null genes. Chronos joint is shown without CNV correction because the published procedure requires multiple cell lines to identify common essential genes and was not applied to this single-line experiment [8]. The DepMap profile was measured separately from the original screen, so cell-line drift is an additional limitation. These results support diagnosing copy-number correction rather than applying it by default.

Additional multivariable and endpoint benchmarks. The following analyses test backward compatibility, multivariable effect concordance, copy-number sensitivity, and preprocessing contracts. They are reported separately because none is a direct test of the primary longitudinal generalization.

7.3 A multi-coefficient design on real data: GSE70038

The benchmarks so far test one coefficient at a time. GSE70038 is the case where the design matrix earns its place: 64,747 guides in 16 libraries from two glioblastoma stem-cell lines and two neural stem-cell lines, with duplicate initial and terminal samples [26]. The scientific target is four cell-line-specific terminal effects estimated *jointly* against one shared initial baseline. Four separate two-group tests cannot express that: each would re-estimate the baseline from its own pair, discarding the other twelve libraries that inform it. This is therefore not a continuous-phenotype benchmark but the multivariable one, and it is the design MAGECKFlute documents as the canonical multi-condition layout.

We followed the design-matrix rules in Table 5 and Box 3 of the MAGECKFlute protocol [27]. The first column is one for every library; each initial library has zeros elsewhere; and each terminal library has one in exactly one of four cell-line columns. The same 16×5 matrix was supplied to beta-binomial regression and the official MAGECK-MLE 0.5.9.5 executable. MAGECK used median normalization and its Wald inference. For the beta-binomial result, guide effects were summarized

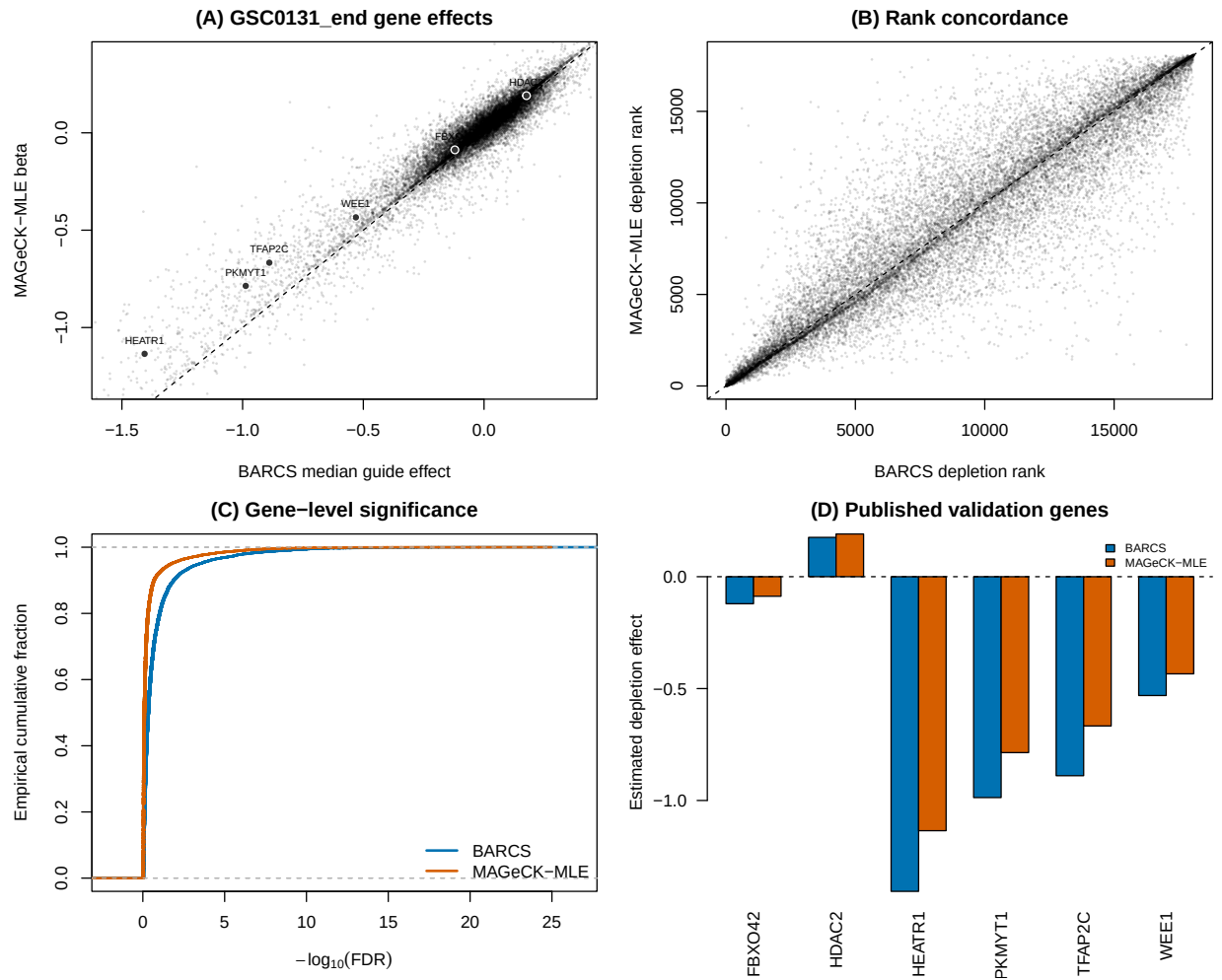
by their within-gene median, and directional guide p -values were combined by Stouffer’s method. This aggregation is an explicit comparison device, not a proposed replacement for MAGeCK’s gene model.

The “effect rank correlation” reported below is calculated in this analysis, not copied from GEO or the MAGeCKFlute paper. Separately for each terminal coefficient, we pair all genes available from both methods. The beta-binomial effect is the within-gene median guide coefficient, and the MAGeCK effect is its official MLE beta. We assign average ranks to these two effect vectors and take their Pearson correlation, which is exactly Spearman’s rank correlation. The 18,077 paired effects and their ranks for every coefficient are written to `results/gse70038/effect_concordance_pairs.csv.gz`.

Supplementary Table S2: Effect concordance between ■ BARCS and ■ official MAGeCK-MLE on GSE70038. Jaccard is the overlap divided by the union of the 200 most depleted genes from each method. Higher agreement is better; column maxima are shown in bold.

Terminal coefficient	Spearman effect correlation	Top-200 Jaccard
GSC0131	0.928	0.550
GSC0827	0.888	0.533
NSCCB660	0.905	0.544
NSCU5	0.915	0.562

Concordance here is the validation rather than the finding. The comparison is not circular: beta-binomial regression conditions on each library total, whereas MAGeCK uses a normalized negative-binomial count model with guide-efficiency estimation [6]. Two models that disagree on the noise distribution, the normalization, and the guide-efficiency treatment nevertheless place the four coefficient vectors in the same order, with all four effect correlations above 0.88. That is the evidence that the model-matrix interface recovers the intended estimand on a real screen rather than a reparameterised approximation of it. These correlations measure agreement of the inferred gene ordering; they are not likelihood, calibration, or predictive-fit statistics and therefore cannot identify a better-fitting method. In GSC0131, the published convergent dependency PKMYT1 has effect -0.987 and FDR 1.87×10^{-8} under beta-binomial regression, versus beta -0.786 and Wald FDR 4.53×10^{-8} under MAGeCK-MLE. Both also recover HEATR1, TFAP2C, and WEE1 depletion. Figure S2 shows the complete GSC0131 comparison; the reproducible PDF contains one page per terminal coefficient.



Supplementary Figure S2: GSE70038 GSC0131 head-to-head. (A) Gene effects, with dark points marking the prespecified genes PKMYT1, HEATR1, TFAP2C, FBXO42, HDAC2, and WEE1. (B) Depletion-rank concordance. (C) Gene-level FDR distributions. (D) Effects for the published validation genes.

7.4 Endpoint benchmark and copy-number audit

To ask which method ranks biological dependencies better, we used the Brunello A375 negative-selection screen of Sanson et al. [28], bundled with CB². This is an ordinary endpoint screen rather than a continuous-phenotype experiment, but it supplies an external target unavailable in GSE70038: reference essential genes should rank ahead of reference nonessential genes [29]. The beta-binomial score combines guide evidence by directional Stouffer aggregation; the MAGeCK score uses the official MLE beta and Wald p -value. Both are oriented so larger scores mean stronger depletion.

7.4.1 Copy-number-aware analysis

Copy number is a gene-level property of A375, not a sample-level covariate: for a given guide it is constant across the plasmid and three terminal libraries. Adding it directly to the four-row sample design would therefore be non-identifiable with the endpoint effect. Instead, we downloaded the A375 gene-level profile (ACH-000219) from DepMap Public 19Q3 [30, 31] and used MAGeCK 0.5.9.5’s official `-cnv-norm` piecewise correction. For an apples-to-apples comparison, the same MAGeCK piecewise normalizer was applied to the beta-binomial within-gene median effects. It adjusts effect sizes after model fitting; it does not refit either count likelihood. The common CNV-complete evaluation set contains 1,506 essential and 869 nonessential genes.

Supplementary Table S3: CNV-aware Sanson A375 benchmark. The final column is Spearman correlation between effect and copy number among reference nonessential genes; values nearer zero indicate less residual CNV-associated depletion. F1 uses nominal FDR 0.05 and a negative effect. Higher discrimination metrics and CNV correlation closest to zero are highlighted.

Method	AUROC	Avg. precision	F1	CNV correlation
■ BARCS	0.9598	0.9786	0.8531	−0.185
■ BARCS + CNV	0.9533	0.9762	0.8531	−0.042
■ Official MAGeCK-MLE	0.9598	0.9795	0.8277	−0.207
■ Official MAGeCK-MLE + CNV	0.9501	0.9762	0.8277	−0.006

CNV correction substantially removes the intended nuisance association: −0.185 to −0.042 for beta-binomial effects and −0.207 to −0.006 for MAGeCK. It does not improve essential-gene discrimination in this screen. AUROC decreases by 0.0066 and 0.0097, respectively, and the 0.05-FDR calls are unchanged. This is not a contradiction: bias removal and gold-standard classification are different targets, and amplified genes can include both dependencies and nondependencies.

After CNV correction, beta-binomial versus MAGeCK global ranking remains indistinguishable: the AUROC difference is 0.00316 (paired stratified-bootstrap 95% CI −0.00265 to 0.00924), and the average precision difference is effectively zero (95% CI −0.00349 to 0.00295). At nominal FDR 0.05, beta-binomial retains 4.18 percentage points more recall (95% CI 2.66 to 5.71) and an F1 advantage of 0.0253 (95% CI 0.0148 to 0.0362).

7.5 External false-discovery benchmark: sensitivity to the analysis contract

A recent Chronos hit-calling preprint compared CB², MAGeCK, JACKS [32], DrugZ [33], and Chronos on single-condition dependency calling and differential viability [34]. The deposited notebooks and intermediate files make this comparison unusually auditable [35]. They expose two distinct observations that should not be conflated. Original CB² has a genuine small-sample power limitation in two-versus-two condition comparisons, but the strongest reported evidence of anti-conservative CB² inference is sensitive to an incompatible preprocessing step.

For guide g and sample j , CB² constructs the library total internally as

$$N_j = \sum_{g \in \mathcal{G}} K_{gj}.$$

The null-only Avana benchmark first restricted the count matrix to selected control genes and then ran CB². Consequently, its beta-binomial denominator became

$$N_j^{\text{subset}} = \sum_{g \in \mathcal{G}_0} K_{gj},$$

rather than the full sequencing depth. Filtering guides is legitimate for evaluation, but it must occur after fitting or be accompanied by immutable full-library totals. This distinction matters because within-sample rescaling leaves guide proportions unchanged while changing the beta-binomial sampling variance through N_j .

We audited this point using the deposited `CB2_no_positives.csv` and `CB2_allgenes.csv` files. The published subset-matrix analysis reports 152 test-negative discoveries at FDR 0.1. Restricting the full-library CB² p -values to exactly the same test genes and reapplying Benjamini–Hochberg gives zero discoveries in all five cell lines (Table S4). The nominal $p < 0.05$ frequency still reaches 14.5% and 11.7% in two lines, so this correction does not establish perfect calibration. It does show that the reported 152 discoveries are not evidence about CB² fitted with its required full-library denominator. A second concern is that the common preprocessing applies Chronos negative-control normalization and rounding before all fits. That transformation is meaningful for Chronos, but it changes the effective sequencing depths of a method whose likelihood treats read totals as sampling information. Equal preprocessing is therefore not automatically equivalent to equal model input.

Supplementary Table S4: Audit of the version-1 Chronos false-discovery benchmark. The Avana row uses the same test genes under two library-total contracts. The PSN1 and DeWeirdt rows are direct BARCS reruns on the deposited normalized count tables; “known” denotes the 28 literature-curated condition–gene combinations used by the preprint, not a complete truth set.

Audit target	■ Reported legacy result	■ BARCS reanalysis
Avana null-only, FDR < 0.1	152 false discoveries after pre-fit guide subsetting	0 after preserving full-library totals
PSN1 two-versus-two null split	0 gene discoveries; conservative gene p -values	4.81% guide $p < 0.05$, 4.65% after control scaling, and 0 gene discoveries
DeWeirdt differential screens	0 discoveries and 0/28 known interactions	38 discoveries and 1/28 known interactions

The differential-screen limitation is real. Original CB² estimates each condition separately and uses the Welch–Satterthwaite degrees of freedom

$$\nu = \frac{(\hat{V}_A + \hat{V}_B)^2}{\hat{V}_A^2/(n_A - 1) + \hat{V}_B^2/(n_B - 1)}.$$

With $n_A = n_B = 2$, $1 \leq \nu \leq 2$. Two independent variance estimates and a heavy-tailed reference distribution leave little power. In contrast, BARCS estimates one guide-level dispersion across the full design. On the PSN1 pseudo-condition split, its guide-level null is close to nominal and its negative-control scale is 1.017. This is evidence that pooled regression repairs part of the finite-sample problem.

It does not solve the full Chronos task. In an exploratory rerun of the ten DeWeirdt comparisons used for the curated-positive analysis [36], BARCS used the four terminal libraries, full count-table totals, a treatment coefficient, guide-level control scaling, Fisher gene aggregation, and BH correction. It recovered MCL1 under A-1331852 in Meljuso, but only one of 28 curated interactions overall. Thirty-seven of its 38 discoveries came from OVCAR8 under olaparib, without recovering BRCA1 or BRCA2. This pattern is consistent with a global difference in screen sensitivity rather than a failure that one additional fixed covariate can remove: with two replicates per arm, a treatment-associated quality shift is nearly collinear with treatment itself. Negative controls that are inactive in both arms also need not represent artifacts affecting strong common-essential knockouts.

The version-1 notebook raises two further reproducibility issues. First, its FDR-calibration helper accepts a requested evaluation gene set but does not use that argument; the calibration curve therefore includes training controls used by Chronos as well as held-out controls. Second, calls intended to remove MAGeCK’s `common` coefficient use a non-mutating `DataFrame.drop()` without assigning the returned object. In the PSN1 calibration, this leaves 34,124 MAGeCK entries versus 17,787 entries for CB², JACKS, DrugZ, and Chronos. These implementation details do not invalidate the biological motivation for empirical-null hit calling, but they make the strongest method-ranking and calibration claims provisional. Similarly, treating thousands of correlated gene–screen indicators as independent observations in a t test yields uncertainty that is narrower than a screen- or cell-line-clustered bootstrap would provide.

The appropriate conclusion is therefore narrower than either “CB² is miscalibrated” or “BARCS solves false discovery.” The regression extension repairs the mean-model and dispersion-pooling limitations, accepts CNV and other identifiable sample covariates, and gives a well-behaved null in the PSN1 check. It does not by itself supply Chronos’s growth dynamics, replicate-quality terms, or empirical differential null. A production differential-screen method still needs immutable library totals, raw-count or likelihood-compatible normalization, dispersion moderation, correlation-aware gene aggregation, held-out control calibration, and blocked replicate-label permutations.